

Hanting Ye

POSTDOC · MARIE-CURIE PHD FELLOW

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Research Interests

MOBILE COMPUTING/SYSTEM, AR/VR/XR, MACHINE LEARNING, HUMAN-ROBOT COLLABORATION, WIRELESS COMMUNICATION

I am passionate about pioneering computing paradigms arising from revolutionary technological changes in the physical world. My doctoral dissertation, titled *Through-Screen Computing*, explores the advent of transparent screens and their growing prevalence in mobile devices. My research primarily explores how to design and implement novel machine learning and signal processing algorithms into these next-generation mobile systems to address the unique challenges transparent screens present to existing applications and to foster innovative new uses. Moving forward, I will focus on investigating *next-generation mobile XR device, smart glasses, -based delivering AI capability to human-robot collaboration and healthcare applications*, e.g., developing remote dual motor and cognitive virtual XR rehabilitation for Parkinson's.

Highlighted Research Achievements

- Designed and developed advanced through-screen Visible Light Communication (VLC) and privacy protection mechanisms for Under-Screen Cameras (USC) and Smartglasses, such as SpiderWeb, VLC Meets USC, Screen Perturbation, NIRF, Unicorn, No Guide, No Cheat, among others. All source codes and collected datasets have been open-sourced in *Through-Screen VLC* and *Screen Perturbation* projects.
- Four papers published at 2*ACM MobiCom, ACM MobiSys, ACM SenSys, all three flagship conferences in the Mobile Computing field under computer system in CSRankings (A*, Acceptance ratio: <25%).
- Smart glasses detection demonstration at ACM HotMobile 2026 received the Best Demo Award. This new detection technology has been showcased and passed by the Duke translation & commercialization office, and one provisional patent was filed in June 2026.
- Led the organization of the Open Lab, E4USA, Demo Days at the I3T Lab and participated in the European Researchers' Night to disseminate research results to the public. The current developed XR rehabilitation application for Parkinson's has been demonstrated at UIUC IMMERSE and will be demonstrated at AWE USA 2026 soon, a top-tier XR and AI industry event. NIRF has also been reported by TU Delft EEMCS department.
- Currently co-supervising 2*PhD students. Had supervised 3*K-12 students, 6*undergraduate student, and 3*master students at Duke and 3*master students in TU Delft. 1*K-12 student has been admitted into Stanford University. 1*undergraduate student at Duke obtained Graduating with Distinction honor and 2*master students in TU Delft obtained 8.5/10 as the final grades. The research outcome of 1*independently supervised master student contributed to the paper accepted at ACM MobiCom'25. The research outcome of 2*supervised undergraduate student contributed to the Best Demo Award at ACM HotMobile'26.
- Lectured on the course "ECE/COMPSCI 356: Computer Network Architecture" at Duke University, and the text book used for this course is the 8th edition of "Computer Networking: A Top-Down Approach" written by James F. Kurose and Keith W. Ross. Gave a lecture "Using Edge Computing to Enhance Foundation Models" for the "ECE/COMPSCI 564: Edge Computing", and worked as the project mentor to guide graduates from proposal writing through the project evaluation process.
- Had been invited to serve on the TPC of ACM MobiSys'26, ACM SenSys'27&26&'25, DATE'27&26, ICCCN'26, EWSN'26&'25 (Distinguished TPC Member), AAAI'25, the Distinguished Reviewer Board of ACM TOSN, and the panelist of the US NSF review panel.
- Heidelberg Laureate Forum (HLF) Young Researcher. HLF is a prestigious event at which 100 carefully selected young researchers in computer science from around the world spend a week interacting with the laureates of these disciplines—recipients of the ACM Turing Award, ACM Prize in Computing, Fields Medal, Abel Prize, and Nevanlinna Prize.
- Doctoral Award with Cum Laude (The highest distinction, top 5%, TU Delft).
- Marie-Curie PhD Fellow. Google scholar citations: 330+; H-index: 10.

Education & Working Experience

Duke University

POSTDOCTORAL RESEARCH ASSOCIATE

Durham, United States

05/2025 – Present

- Advisor: [Prof. Maria Gorlatova](#)
- Intelligent Interactive Internet of Things (I3T) Lab, Department of Electrical and Computer Engineering.

Delft University of Technology (TU Delft)

POSTDOCTORAL RESEARCH ASSOCIATE

Delft, Netherlands

10/2024 – 04/2025

- Advisor: [Prof. Qing Wang](#)
- Embedded Systems (ES) Group, Computer Science Department.

Delft University of Technology (TU Delft)

PHD WITH Cum Laude (MARIE-CURIE PHD FELLOW)

Delft, Netherlands

10/2020 – 09/2024

- Thesis: *Through-Screen Computing*
- Promoters: [Prof. Qing Wang](#), [Prof. Marco Antonio Zúñiga Zamalloa](#)
- Embedded Systems (ES) Group, Computer Science Department.

PureLiFi Technology

VISITING PHD STUDENT

- Advisor: [Prof. Harald Haas](#)
- [pureLiFi](#) is the world leader in Light Fidelity (LiFi) innovation.

Edinburgh, United Kingdom

09/2022 - 01/2023

University of Electronic Science and Technology of China (UESTC)

M.S. IN ELECTRONICS AND COMMUNICATION ENGINEERING

- Advisor: [Prof. Xin Kang](#)
- National Key Laboratory of Wireless Communication

Chengdu, China

09/2017 - 06/2020

Chongqing University (CQU)

B.S. IN COMMUNICATION ENGINEERING WITH HONORS

- Hongshen Honors School & School of Microelectronics and Communication Engineering

Chongqing, China

09/2013 - 06/2017

Fellowships & Grants

For Research Activities

NSF Ideas Lab: Breaking the Low Latency Barrier for Verticals in Next-G Wireless Networks (Breaking Low), Lead of I3T lab

US, END-TO-END DELIVERY TECHNOLOGY FOR INTERACTIVE MULTI-PERSON XR REHABILITATION ACTIVITIES

≈743 K\$

10/2025

NSF Future CoRe, Proposal Writing

US, PROJECT TITLE: HCC: FOUNDATIONS OF AUTOMATED EVALUATION AND MONITORING OF AR GUIDANCE IN PERVASIVE XR AND HUMAN-ROBOT COLLABORATION (HRC) APPLICATIONS

TBD

02/2026

Meta, PI, Proposal Writing

US, PROJECT ARIA GEN 2

4 glasses

12/2025

FCAT, Senior Personnel, Proposal Writing

US, PROJECT TITLE: HYBRID NERF-3DGS FOR HIGH-FIDELITY MOBILE XR 3D RECONSTRUCTION

≈100 K€

06/2025

HORIZON-MSCA-DN-2024-PRJ, Proposal Writing

EU, PROJECT TITLE: ELEPHANT: LARGE AI MODELS IN 6G NETWORKS

≈200 K€

06/2026-12/2029

NWO, Open Competition Domain Science - XS, Proposal Writing

THE NETHERLANDS, SUBMITTED, PROJECT TITLE: COVERING YOUR CAMERA WITH TRANSPARENT SCREEN!

≈50 K€

03/2024

H2020-MSCA-ITN-2018, Marie-Curie Early Stage Researcher

EU, PROJECT TITLE: ENLIGHT'EM-EUROPEAN TRAINING NETWORK IN LOW-ENERGY VISIBLE LIGHT IoT SYSTEMS

≈128 K€

06/2019-12/2023

Abbe grant

CARL-ZEISS-STIFTUNG

2000 €

2024

Heidelberg Laureate Forum (HLF) young researcher travel grant

HEIDELBERG LAUREATE FORUM FOUNDATION (HLFF)

1500 €

2024

EWSN student travel grant

TECHNOLOGY INNOVATION INSTITUTE

1000 €

2024

ACM MobiSys student travel grant

ACM SIGMOBILE COMMUNITY

1500 €

2023

ACM SenSys student travel grant

ACM SIGMOBILE COMMUNITY

1000 €

2021

IEEE INFOCOM student travel grant

IEEE ComSoc COMMUNITY

1000 €

2021

For Academic Activities

(Top 1%) The 1st Class Master Degree Scholarship

CHINA, UNIVERSITY OF ELECTRONIC SCIENCE AND TECHNOLOGY OF CHINA

2000 €

2018, 2019

(Top 3%) Enrollment Scholarship for Master Student

CHINA, UNIVERSITY OF ELECTRONIC SCIENCE AND TECHNOLOGY OF CHINA

1000 €

2017

(Top 1%) National Scholarship

CHINA, CHONGQING UNIVERSITY

1000 €

2016

(Top 3%) National Inspirational Scholarship

CHINA, CHONGQING UNIVERSITY

600 €

2015

(Top 5%) The 2nd Class Bachelor Degree Scholarship

CHINA, CHONGQING UNIVERSITY

400 €

2014, 2015

Honors & Awards

2026	ACM HotMobile Best Demo Award
2025	(Top 5%) Doctoral Award with Cum Laude of TU Delft
2024	(100 total in computer science per year) Heidelberg Laureate Forum Young Researcher
2019	(Top 1%) Excellent student leader Awards of UESTC
2018, 2019	(Top 1%) Outstanding Student Awards of UESTC
2017	(Top 0.5%) Outstanding Graduates Awards of Chongqing province
2017	(Top 1%) Outstanding Graduates Awards of CQU
2016	(Top 1%) Outstanding Student Awards of CQU

Competition Activities

2018	15th China Graduate Mathematical Contest in Modelling	(Top 1%) 1st Prize
2016	International Mathematical Contest in Modeling	(Top 5%) Honorable
2015	China Undergraduate Mathematical Contest in Modelling	(Top 1%) 1st Prize
2015	Electronic Skills Competition of Chongqing University	(Top 5%) Honorable
2014	Mobile app design competition of Chongqing University	(Top 1%) 1st Prize

Publications

[†] PhD/Master students (co-)advised by me; [Google Scholar](#) Citations: 330+; H-index: 10

[1] No Guide, No Cheat: Detecting Smart Glasses via AR Optical Signatures

ACM HotMobile

HANTING YE, TIANYI HU, MARIA GORLATOVA.

2026

- The International Workshop on Mobile Computing Systems and Applications.
- (A conference)

[2] SHARE: Towards Head-Mounted AR with User-Centric SLAM in Shared Human-Robot Workspaces

ACM IMWUT

TIANYUAN DU[†], TIANYI HU, HANTING YE, MARIA GORLATOVA.

2026

- The Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies.
- (A* conference)

[3] ARTOO-DARTU: Studying AR-Assisted Human-Robot Collaboration with AR Obstruction Mitigation during a Warehouse Task

ACM MobileHCI

CHRISTIAN FRONK[†], HANTING YE, ZHEHAN QU, MARIA GORLATOVA.

2026

- ACM International Conference on Mobile Human-Computer Interaction.

[4] fARfetch: Enabling Collocated AR-HRC in Large Visually Diverse Environments with VLM-Driven AR Content Adaptation

IEEE RO-MAN

CHRISTIAN FRONK[†], DAVID HUNT, HANTING YE, MIROSLAV PAJIC, MARIA GORLATOVA.

2026

- IEEE International Conference on Robot and Human Interactive Communication.

[5] NIRF: Detecting Cameras That Hide Behind Screen

ACM MobiCom

HANTING YE, NIELS VAN DER KOLK[†], QING WANG.

2025

- ACM Conference on Mobile Computing and Networking.
- (Acceptance ratio: 17.1%, A* conference)

[6] Computing behind Transparent Screen

EWSN

HANTING YE, QING WANG.

2024

- International Conference on Embedded Wireless Systems and Networks.
- (Acceptance ratio: 22.2%)

[7] Screen Perturbation: Adversarial Attack and Defense on Under-Screen Camera

ACM MobiCom

HANTING YE, GUOHAO LAN, JINYUAN JIA, QING WANG.

2023

- ACM Conference on Mobile Computing and Networking.
- (Acceptance ratio: 24.4%, A* conference)

[8] When VLC Meets Under Screen Camera

ACM MobiSys

HANTING YE, JIE XIONG, QING WANG.

2023

- ACM Conference on Mobile Systems, Applications, and Services.
- (Acceptance ratio: 20.7%, A* conference)

[9] SpiderWeb: Enabling Through-Screen Visible Light Communication

ACM SenSys

HANTING YE, QING WANG.

2021

- ACM Conference on Embedded Networked Sensor Systems.
- (Acceptance ratio: 18%, A* conference)

[10] Optimization for Full-Duplex Rotary-Wing UAV-Enabled Wireless-Powered IoT Networks

IEEE TWC

HANTING YE, XIN KANG, JINGON JOUNG, YING-CHANG LIANG.

2020

- IEEE Transactions on Wireless Communications
- A* Journal

[11] Fingertip Air-Writing with Ambient Light

EAI MobiQuitous

HAO LIU[†], HANTING YE, XIANGRUI ZHANG, JIE YANG, QING WANG.

2023

- EAI Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services.

[12] Through-Screen Visible Light Sensing Empowered by Embedded Deep Learning

AIChallengesIoT

HAO LIU[†], HANTING YE, JIE YANG, QING WANG.

2021

- ACM SenSys Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things.

[13] Joint Uplink-and-Downlink Optimization of 3D UAV Swarm Deployment for Wireless-Powered IoT Networks

IEEE IoT-J

HANTING YE, XIN KANG, JINGON JOUNG, YING-CHANG LIANG.

2021

- IEEE Internet of Things Journal

[14] Optimization for Wireless-Powered IoT Networks Enabled by an Energy-Limited UAV Under Practical Energy Consumption Model

IEEE WCL

HANTING YE, XIN KANG, JINGON JOUNG, YING-CHANG LIANG.

2021

- IEEE Wireless Communications Letters

[15] Cognitive Backscatter NOMA Networks With Multi-Slot Energy Causality

IEEE CL

JIE WANG*, HANTING YE*, XIN KANG, SUMEI SUN, YING-CHANG LIANG.

2020

- IEEE Communications Letters
- (*Equal contribution)

[16] Joint Uplink and Downlink 3D Optimization of an UAV Swarms for Wireless-powered NB-IoT

IEEE GLOBECOM'19

HANTING YE, XIN KANG, JINGON JOUNG, YING-CHANG LIANG.

2019

- IEEE Global Communications Conference

[17] Optimal Time Allocation for Full-duplex Wireless-Powered IoT Networks with UAV

IEEE ICC'19

HANTING YE, XIN KANG, JINGON JOUNG, YING-CHANG LIANG.

2019

- IEEE International Conference on Communications

[18] Full-duplex Wireless-powered IoT Networks with Unmanned Aerial Vehicle

IEEE ICTC'18

HANTING YE, XIN KANG, YING-CHANG LIANG, JINGON JOUNG.

2018

- IEEE International Conference on ICT Convergence
- Invited paper

Demonstrations

[†] PhD/Master students (co-)advised by me;

[1] Demo: WordPlay Party: Collaborative XR game for Parkinson's Rehab

AWE USA

HANTING YE, TIANYUAN DU[†], JAMES FINLEY, JIASI CHEN, MARIA GORLATOVA.

June 2026

- Augmented World Expo USA.
- Long Beach, California

[2] Demo: WordPlay Party: Towards Collaborative XR Application Latency Guarantees for Parkinson's Rehabilitation

UIUC IMMERSE

HANTING YE, TIANYUAN DU[†], JAMES YONG[†], JIACHENG GUO[†], JAMES FINLEY, JIASI CHEN, MARIA GORLATOVA.

May 2026

- UIUC IMMERSE Symposium.
- Urbana, Illinois

[3] Demo: Catch Smart Glasses If You Scan: Robot-Controlled vs. UI-Guided

ACM HotMobile

HANTING YE, TIANYI HU, VALIANTIN KASABRUKHAU[†], SULAIMAN KHYDYR UULU[†], MARIA GORLATOVA.

February 2026

- The International Workshop on Mobile Computing Systems and Applications.
- (A conference, *Best Demo Award*)

[4] Demo: Control the ARmada: LMM Coordination for Multi-Robot AR-HRC

ACM HotMobile

CHRISTIAN FRONK[†], HANTING YE, MIROSLAV PAJIC, MARIA GORLATOVA.

February 2026

- The International Workshop on Mobile Computing Systems and Applications.
- (A conference)

Patents

[1] System and Method for Detecting the Presence of Smart Glasses

Provisional Patent

HANTING YE, MARIA GORLATOVA.

June 02 2026

- Provisional patent filed June 2, 2026.

[2] Methods for Detecting Smart Glasses with Waveguide-based AR Displays

Disclosure

HANTING YE, MARIA GORLATOVA.

February 23 2026

- Duke University invention disclosure form filed February 23, 2026.

Research Project

Breaking Low - End-to-End Delivery Technology for Interactive Multi-person XR Rehabilitation Activities

Duke

POSTDOCTORAL RESEARCH ASSOCIATE

- Leading a multidisciplinary team, including game developers and O-RAN engineers, in developing a multi-person XR rehabilitation application on commercial XR platforms (Meta Quest 3 and VIVE Ultimate Trackers) for Parkinson's patients. The remote application provides dual-task training for individuals with both motor and cognitive impairments, enabling at-home rehabilitation under remote therapist supervision. A co-located multi-user demo of the *WordPlay* application has been released.
- Coordinating with academic (University of Michigan) and industry (Nokia) partners in the Breaking Low project to build ultra-low-latency networking infrastructure on O-RAN systems that support immersive XR experiences for Parkinson's patients using *WordPlay*.
- Collaborating with clinical partners at the University of Southern California to develop a patient tracking data collection pipeline and to recruit Parkinson's patients for an IRB-approved user study to begin real-world evaluation.

AR for Human-Robot Collaboration

Duke

POSTDOCTORAL RESEARCH ASSOCIATE

- Co-advising a PhD student on developing edge-assisted systems that allow AR content to display robot internal states without obstructing important real-world objects. Also guiding efforts to enhance the legibility of AR content in large-scale and complex environments as part of DARPA-funded projects.
- Co-advising a PhD student on designing new approaches to reduce communication and computation overhead in existing multi-agent SLAM systems in order to meet strict latency constraints for AR visual cues. These cues provide spatial awareness to heterogeneous agent groups (humans and robots) in human-robot collaboration contexts.
- **Research Output: 1 Paper @ IMWUT'26; 1 Paper @ MobileHCI'26, 1 Paper @ RO-MAN'26.**

Privacy Protection of Under-Screen Camera

TU Delft

PHD STUDENT

- Developed an optical model to simulate various screen pixel layouts under active light interference, assessing effects on pixel size, shape, and arrangement for under-screen cameras.
- Innovated and implemented 'One-Pixel Perturbation' and 'Multi-Pixel Perturbation' strategies, achieving over 90% reduction in neural network recognition accuracy for images, thus significantly enhancing key privacy safeguards for sensitive information like facial data.
- Led the conceptualization and mentoring of student projects focused on using retroreflections and time-of-flight sensors for the detection of hidden under-screen cameras, fostering practical skills and research methodologies.
- **Research Output: 1 Paper @ MobiCom'23; 1 Paper @ MobiCom'25.**

Enabling Through-Screen Visible Light Communication (VLC)

TU Delft

PHD STUDENT & VISITING RESEARCHER AT PURELIFI

- Pioneered the concept of through-screen VLC, demonstrating its practical feasibility and laying the groundwork for future advancements in optical communications.
- Identified and analyzed a color-pulling effect in transparent OLED screens impacting color-based CSK signals; developed a novel pixel-sweeping algorithm to mitigate rolling-shutter effects in under-screen cameras.
- Designed a spiderweb-inspired 'SWebCSK' modulation scheme, integrating a color-based approach with a unique slope-boosting demodulation technique, significantly enhancing signal clarity and transmission efficiency.
- **Research Output: 1 Paper @ MobiSys'23; 1 Paper @ SenSys'21.**

Embedded Deep Learning Empowered Fingertip Air-Writing with Ambient Light

TU Delft

CO-ADVISOR OF MSc STUDENT

- Innovatively designed and prototyped an ambient light-based fingertip air-writing system.
- Conducted comprehensive analysis of in-air digit writing patterns, leading to the creation of a unique dataset for air-written digits.
- Developed a novel calibration method to significantly improve the system's robustness and accuracy in diverse lighting conditions.
- Designed two advanced embedded deep learning models to decode spatial and temporal patterns in dynamic shadows.
- **Research Output: 1 Paper @ AIChallengeIoT'21; 1 Paper @ MobiQuitous'23.**

Resource Allocation and Optimization of Time/Power-Limited UAV Networks

UESTC

MASTER STUDENT

- Devised an innovative time allocation algorithm to optimize system performance under sparse user distribution scenarios, leveraging the interplay between UAV hovering and flight times for enhanced efficiency.
- Designed a comprehensive framework for dense user environments, incorporating joint optimization of 3D UAV positioning, device-UAV associations, and scheduling for both uplink and downlink, significantly boosting network throughput and reliability.
- **Research Output: 1 Paper @ TWC; 1 Paper @ IoT-J; 1 Paper @ WCL; 1 Paper @ CL; 1 Paper @ GLOBECOM; 1 Paper @ ICC; 1 Paper @ GLOBECOM; 1 Paper @ ICTC.**

Research on Key Technologies of Passive IoT and Cellular System Convergence

UESTC

MASTER STUDENT

- Conducted in-depth analysis on channel estimation and signal detection performance for an OFDM-based Ambient Backscatter system within a 3GPP time-varying channel, contributing to advancements in understanding and optimizing backscatter communication technologies.
- Developed a novel pilot policy tailored for transmitters and backscatter devices, effectively minimizing pilot overhead in scenarios with a high volume of backscatter devices, thereby enhancing system scalability and efficiency.

Optical Transport Network and Network Value Evaluation

UESTC

MASTER STUDENT (NATIONAL MATHEMATICAL MODELING CONTEST)

- Developed comprehensive single-span and multi-span transmission models for optical links.
- Innovated a network value evaluation model, effectively solving it using a genetic algorithm.
- Designed an unequal probability QAM method, boosting system performance and transmission reliability in optical transport networks.

Undergraduate Thesis on 802.11ac Standard WLAN Synchronization System

CQU

UNDERGRADUATE STUDENT

- Implemented and analyzed various OFDM receiver synchronization algorithms including S & C, Minn, Park, and MNC for the 802.11ac WLAN standard. This involved in-depth coding, testing, and optimization to ensure accurate synchronization in high-speed wireless networks.

Teaching Experience

ECE/COMPSCI 564: *Project Mentor, Edge Computing* [5 credits], MSc course, Spring, 2026

ECE/COMPSCI 356: *Lecturer, Computer Network Architecture* [5 credits], BSc course, Winter, 2025

CSE2425: *Teaching assistant, Embedded Software* [5 credits], BSc course, Q2, 2020, 2021, 2022, 2023

CESE4110: *Teaching assistant, Visible Light Communication and Sensing* [5 credits], MSc course, Q4, 2021-2022, Q1, 2023-2024

Selected Research Supervision

Tianyuan Du, PhD Student, Low-Latency Guarantees in AR Human-Robot Collaboration and Networked AR Healthcare Systems

Duke

I AM HIS CO-SUPERVISOR

05/2025–Present

- I mainly guided the student in identifying that AR users exhibit varying sensitivity to virtual content and robot tracking in human-robot collaborative scenarios. He developed an adaptive algorithm to address imbalanced communication and computation resource allocation across heterogeneous agents (humans and robots), enabling the edge-based SLAM system to maintain spatial positioning within a strict 20ms latency constraint. This work has been submitted to **ACM IMWUT**. I am also leading him in investigating cross-XR applications and O-RAN multi-user Parkinson's rehabilitation system design to guarantee low user-to-user latency in **NSF Breaking Low project**.

Christian Fronk, PhD Student, Enhancing the Safety of AR Content in Human-Robot Collaboration (HRC) Scenarios

Duke

I AM HIS CO-SUPERVISOR

05/2025–Present

- I mainly guided the student in improving the visibility of critical real-world objects and the legibility of AR content within the user's field of view in HRC scenarios. These works focused on ensuring that virtual displays, such as those visualizing the robot's internal state via an edge-computing-supported object detection pipeline, do not obscure key physical elements (e.g., safety-related signage in a warehouse). Semantic segmentation and vision-language model capabilities are also leveraged to enhance the clarity of virtual content, such as robot target locations, in outdoor and complex environments. These works are submitted to **IEEE RO-MAN** and **ACM MobileHCI**.

Valiantsin Kasabrukha, Undergraduate Student, Robot-arm-based, controllable scanning platform with user-customizable trajectories

Duke

I AM HIS CO-SUPERVISOR

12/2025–Present

- I mainly guided the student in leveraging the robot arm to enable a reproducible smart glasses scanning process, where scanning trajectories can be customized by users. The scanning platform consists of a 4-DoF robot (PincherX 100 from Trossen Robotics), a 3D-printed face mask for mounting the smart glasses, and a smartphone camera on a stand. For robot control, we use a Linux-based Beelink Mini S12 running ROS2. We developed a UI with 11 predefined basic robot arm movements, which can be combined and adjusted in both order and amplitude of each movement to mimic human head motions within ranges of 140° yaw, 120° pitch, and 80°. These works are submitted to **ACM HotMobile** and received **Best Demo Award**.

Sulaiman Khydyr uulu, Undergraduate Student, Smartphone-based scanning pipeline with UI indicators

Duke

I AM HIS CO-SUPERVISOR

12/2025–Present

- I mainly guided the student in developing a detector app on a Google Pixel 3 with UI indicators to guide users during scanning, ensuring that the smart glasses are observed from a comprehensive range of angles. We deploy the Google ML Kit face detection model to identify facial landmarks and use them to calculate the spatial orientation of the user's head pose. To simplify the user guidance experience, we create a ring around the user composed of 80 short grey bars. These bars represent different facial orientations. When a grey bar is covered for a certain period of time, it turns green. We also display instructional text at the bottom of the app to guide users to move toward any uncovered orientations until all the grey bars turn green. These works are submitted to **ACM HotMobile** and received **Best Demo Award**.

Adrian Willet, K-12 Student, Gaussian Streaming: Visualizing Worlds on Mobile XR Devices

Duke

I SUPERVISED HIM INDEPENDENTLY

06/2025–Present

- I mainly supervise him to work on developing a new Gaussian Splatting technique to help mobile XR device, e.g., Meta Quest 3, to load reconstructed 3D scenes. The final supervision result is showed at North Carolina School of Science and Mathematics (NCSSM) Summer Research & Innovation Program Presentation. He is now admitted to **University of Toronto**.

Junfeng Lin, K-12 Student, Investigating the Robustness of Open-Vocabulary Models in AR

Duke

I AM HIS CO-SUPERVISOR

07/2025–Present

- Supervised him in exploring the robustness of state-of-the-art object detection models, including open-vocabulary approaches (e.g., GroundingDINO) and conventional models (e.g., YOLOE) in AR contexts. Guided the analysis of key performance constraints, such as variations in user speech prompts and environmental factors (ambient lighting, shooting angle, distance), and their impact on detection accuracy. He is now admitted to **Stanford University**.

Dhun Pandya, K-12 Student, Machine Learning Model Evaluation for SLAM Error Prediction

Duke

I AM HER CO-SUPERVISOR

06/2025–07/2025

- I guided her in implementing and evaluating machine learning models (e.g., CNNs, MLPs, random forest, regression) to analyze SLAM error, while exploring how model inputs and training parameters affect runtime and accuracy. The final supervision result is showed at North Carolina School of Science and Mathematics (NCSSM) Summer Research & Innovation Program Presentation. He is now admitted to **UNC**.

Niels van der Kolk, Master Student, Under-Screen Camera Detection

TU Delft

I SUPERVISED HIM INDEPENDENTLY

06/2022–07/2023

- Under my supervision, he has developed the world's first under-screen camera detection system using a ToF sensor. This innovative project addresses growing concerns about the potential privacy risks posed by hidden spy cameras, highlighting our commitment to leveraging technology for societal well-being. The related work contributed to **ACM MobiCom**. His final grade is **8.5/10**.

Tolga Parlan, Master Student, Coloring the Pulse Position Modulation in VLC

TU Delft

I AM HIS CO-SUPERVISOR

11/2021–06/2023

- Under my co-supervision (together with Prof. Q. Wang), he successfully design and implement a new modulation scheme to improve VLC system performance. His final grade is **7.5/10**.

- He is now a PhD student in TU Delft. Under my co-supervision (together with Prof. Q. Wang and Prof. J. Yang), he has published a paper at **ACM AIChallengeloT@SenSys**, and a paper at **EAI MobiQuitous**. His final grade is **8.5/10**.

Professional Service

National funding agency service

- US National Science Foundation (NSF): proposal review panelist, communications and networking, 2025

TPCs for Conferences

- The 25th ACM Conference on Embedded Networked Sensor Systems (SenSys'27)
- The 30th Design, Automation and Test in Europe Conference (DATE'27)
- The 24th ACM Conference on Mobile Systems, Applications, and Services (MobiSys'26)
- The 24th ACM Conference on Embedded Networked Sensor Systems (SenSys'26)
- The 23rd International Conference on Embedded Wireless Systems and Networks (EWSN'26)
- The 29th Design, Automation and Test in Europe Conference (DATE'26)
- The 35th International Conference on Computer Communications and Networks (ICCCN 2026)
- The 23rd ACM Conference on Embedded Networked Sensor Systems (SenSys'25)
- The 22nd International Conference on Embedded Wireless Systems and Networks (EWSN'25, **Distinguished TPC Member**)
- The 39th Annual AAAI Conference on Artificial Intelligence (AAAI'25)
- The second Workshop on Enhancing Security, Privacy, and Trust in Extended Reality (XR) Systems (XRSecurity'26)
- The first Workshop on Enhancing Security, Privacy, and Trust in Extended Reality (XR) Systems (XRSecurity'25)

Organizing Committee

- Web Co-Chair, EWSN'25
- Local Arrangements Co-Chair, EWSN'21
- Student Volunteer, IEEE ICNP'21
- Student Volunteer, IEEE ICCS'18

Reviewers for Conferences

- The 40th Annual Conference on Neural Information Processing Systems (NeurIPS'26)
- The 2026 ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT'26)
- The 2025 ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT'26)
- The 32th IEEE Conference on Virtual Reality and 3D User Interfaces (VR'26)
- The 40th Annual Conference on Neural Information Processing Systems (NeurIPS'26)
- The 18th European Conference on Computer Vision (ECCV'26)
- The 2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR'26)
- The 24th IEEE International Symposium on Mixed and Augmented Reality (ISMAR'25)
- The 2025 IEEE/CVF International Conference on Computer Vision (ICCV'25)
- The 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR'25)
- The 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR'24)

Reviewers for Journals

- ACM Transactions on Sensor Networks (TOSN, **Distinguished Reviewer Board**)
- IEEE Journal on Selected Areas in Communications (JSAC)
- IEEE Transactions on Wireless Communications (TWC)
- IEEE Transactions on Mobile Computing (TMC)
- IEEE Transactions on Services Computing (TSC)
- IEEE Transactions on Communications (TCOM)
- IEEE Internet of Things Journal (IoT-J)
- IEEE Communications Letters (CL)
- IEEE Wireless Communications Letters (WCL)

(External) Reviewers for Conferences and Journals

- 2026: ACM MobiCom
- 2025: IEEE INFOCOM, ACM MobiSys
- 2024: IEEE INFOCOM, ACM MobiCom
- 2023: IEEE INFOCOM, IEEE SECON
- 2022: IEEE INFOCOM, ACM SenSys
- 2021: IEEE INFOCOM
- 2017-2020: IEEE TWC, IEEE TCOM, IEEE TVT, IEEE IoT-J

Lecturers & Presentations & Talks

- **Lecturer: 02/2026, Durham, USA, Title:** Using Edge Computing to Enhance Foundation Models.
- **Conference Presentation: 02/2026, Atlanta, USA, Title:** No Guide, No Cheat: Detecting Smart Glasses via AR Optical Signatures.
- **Conference Presentation: 11/2025, Hong Kong, China, Title:** NIRF: Detecting Cameras That Hide Behind Screen.
- **Conference Presentation: 12/2024, Abu Dhabi, United Arab Emirates, Title:** Computing behind Transparent Screen.
- **Conference Presentation: 10/2023, Madrid, Spain, Title:** Screen Perturbation: Adversarial Attack and Defense on Under-Screen Camera.
- **Conference Presentation: 06/2023, Helsinki, Finland, Title:** When VLC Meets Under-Screen Camera.
- **Talk: 06/2023, Soesterberg, The Netherlands, Title:** When VLC Meets Under-Screen Camera.
- **Talk: 10/2022, Linz, Austria, Title:** Through-Screen LiFi System.
- **European Researchers' Night: 10/2022, Vienna, Austria, Title:** Can LEDs talk?
- **Conference Presentation: 11/2021, Coimbra, Portugal, Title:** SpiderWeb: Enabling Through-Screen Visible Light Communication.
- **Conference Presentation: 12/2019, Hawaii, USA, Title:** Joint Uplink and Downlink 3D Optimization of an UAV Swarms for WP NB-IoT.
- **Conference Presentation: 05/2019, Shanghai, China, Title:** Optimal Time Allocation for Full-duplex Wireless-Powered IoT Networks with UAV.

News List

- **EEMCS Department of TU Delft: 11/2025, LinkedIn Post [Link](#):** *TU Delft's Embedded AI Lab for developing a new system that can detect behind-screen hidden cameras!*